

Roots (Part III) Homework

Name
Date
Course

Simplify.

Answers

① $\sqrt[3]{729}$

9

② $\sqrt[3]{-1000}$

-10

③ $\sqrt[3]{1}$

1

④ $\sqrt[3]{56}$

$2\sqrt[3]{7}$

⑤ $\sqrt[3]{-88}$

$-2\sqrt[3]{11}$

⑥ $\sqrt[3]{270}$

$3\sqrt[3]{10}$

⑦ $\sqrt[3]{x^{15}}$

x^5

⑧ $\sqrt[3]{x^{18}}$

x^6

⑨ $\sqrt[3]{x^{30}}$

x^{10}

⑩ $\sqrt[3]{4000x^6}$

$10x^2\sqrt[3]{4}$

⑪ $\sqrt[3]{135x^7}$

$3x^2\sqrt[3]{5x}$

⑫ $\sqrt[3]{320x^4}$

$4x\sqrt[3]{5x}$

⑬ $\sqrt[3]{-27x^5}$

$-3x\sqrt[3]{x^2}$

⑭ $\sqrt[3]{108x^{12}}$

$3x^4\sqrt[3]{4}$

⑮ $\sqrt[3]{5x} \cdot \sqrt[3]{50x^2}$

$5x\sqrt[3]{2}$

⑯ $\sqrt[3]{9x^2} \cdot \sqrt[3]{6x^2}$

$3x\sqrt[3]{2x}$

⑰ $\sqrt[3]{4x^5}(\sqrt[3]{4x})$

$2x^2\sqrt[3]{2}$

⑱ $\sqrt[3]{15x^8} \cdot \sqrt[3]{25x^2}$

$5x^3\sqrt[3]{3x}$

⑲ $\sqrt[3]{16x^3}(\sqrt[3]{10x^{12}})$

$2x^5\sqrt[3]{20}$

⑳ $8\sqrt[3]{24x^{11}} - 5x^2\sqrt[3]{81x^5}$

$x^3\sqrt[3]{3x^2}$

㉑ $-2\sqrt[3]{40x^7} + 10x^2\sqrt[3]{5x}$

$6x^2\sqrt[3]{5x}$

㉒ $\frac{8}{\sqrt[3]{4}}$

$\frac{8\sqrt[3]{2}}{2} = 4\sqrt[3]{2}$

㉓ $\frac{-7}{\sqrt[3]{32}}$

$\frac{-7\sqrt[3]{2}}{4}$

㉔ $\sqrt[4]{1}$

1

㉕ $\sqrt[4]{256}$

4

㉖ $\sqrt[4]{x^{24}}$

x^6

㉗ $\sqrt[4]{x^{36}}$

x^9

㉘ $\sqrt[4]{x^9}$

$x^2\sqrt[4]{x}$

㉙ $\sqrt[4]{x^{18}}$

$x^4\sqrt[4]{x^2}$

㉚ $\sqrt[4]{\frac{9}{16}}$

$\frac{\sqrt[4]{9}}{2}$

$$\textcircled{31} \sqrt[4]{\frac{81}{2}}$$

$$\boxed{\frac{3\sqrt[4]{8}}{2}}$$

$$\textcircled{32} \sqrt[4]{\frac{2}{9}}$$

$$\boxed{\frac{\sqrt[4]{18}}{3}}$$

$$\textcircled{33} \sqrt[3]{\frac{2}{9}}$$

$$\boxed{\frac{\sqrt[3]{6}}{3}}$$

$$\textcircled{34} \sqrt[3]{\frac{9}{128}}$$

$$\boxed{\frac{\sqrt[3]{36}}{8}}$$

$$\textcircled{35} \sqrt[3]{\frac{56x^9}{9}}$$

$$\boxed{\frac{2x^3\sqrt[3]{21}}{3}}$$

$$\textcircled{36} \sqrt[4]{\frac{32x^{10}}{405}}$$

$$\boxed{\frac{2x^2\sqrt[4]{250x^2}}{15}}$$

$$\textcircled{37} \sqrt[4]{\frac{3}{162x^5}}$$

$$\boxed{\frac{\sqrt[4]{24x^3}}{6x^2}}$$